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Mechanisms of efficient heat transfer in n-octadecane/silica phase change microcapsules: A molecular dynamics study

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ABSTRACT

In this paper, a paraffin (n-octadecane) / silicon dioxide (SiO2) three-layer model was constructed. Langevin temperature control method was used to set heat sources at different temperatures, and the effects of different temperature differences and different core/shell ratios on heat transfer were simulated respectively. It is found that in the initial stage of heat transfer, the influence of temperature difference on the temperature rise of C18H38 layer is not obvious, but with the progress of heat transfer, its influence is gradually enhanced. The reduction of temperature difference will lead to a decrease in the heat exchange rate. The excellent thermal conductivity of the shell allows it to rapidly attain the temperature of the heat source, thereby facilitating swift heat transfer to the internal phase change material (PCM). The heating rate of the PCM is directly proportional to the temperature differential between the shell and its external environment. In addition, the change of shell thickness has a certain influence on the heating rate of PCM. Although the increase of core/shell ratio can slightly increase the heating rate of C18H38 layer and optimize the temperature distribution, the shell thickness is not the main factor affecting the thermal resistance of the whole phase change microcapsule because the thermal conductivity of the shell material is much higher than that of PCM.

1. Introduction

In the current wave of research on sustainable energy development and renewable use, phase change materials stand out due to their unique thermal management properties. Such materials can efficiently absorb or release heat near the phase transition temperature, achieving thermostatic control over a specific temperature range. They have the advantages of high energy density, long life, good chemical stability and almost isothermal energy storage and release [1–3]. It has shown great application potential in the field of thermal energy management and has become a hot research direction.

Phase change materials (PCM) usually have low thermal conductivity and specific heat capacity, which limits their application in heat storage. To improve performance, researchers have tried doping nanoparticles [4]. Zhao et al. [5] found through experiments and simulations that the addition of modified copper oxide nanoparticles could significantly improve the thermal conductivity of n-octadecane, and the addition of surfactant further optimized the thermal physical properties. Yan et al. [6] pointed out that the addition of carbon nanotubes can significantly increase the thermal conductivity of erythritol, especially when the axial length of carbon nanotubes is less than the mean free path of phonons, and its random distribution can improve the anisotropy of thermal conductivity. Yu et al. [7] proposed a Na2CO3/ Graphene-based PCM. Through molecular dynamics simulation, it was found that the intercalation structure could limit the ion motion, decrease the self-diffusion coefficient, increase the shear viscosity, and significantly increase the thermal conductivity.

Although phase-change energy storage technology can realize energy storage and environmental temperature regulation with the help of phase-change materials, it still faces many challenges in large-scale applications, such as leakage, corrosion, lack of stability and poor heat transfer performance, which seriously restricts its wide application in various fields. Therefore, phase change microcapsule technology came into being and became an effective way to solve the above problems. In this technology, polymer materials are usually used as the shell, and phase change materials are used as the core component. The core material is coated with solid particles forming a core-shell structure [8]. It can not only increase the stability of phase change materials, but also

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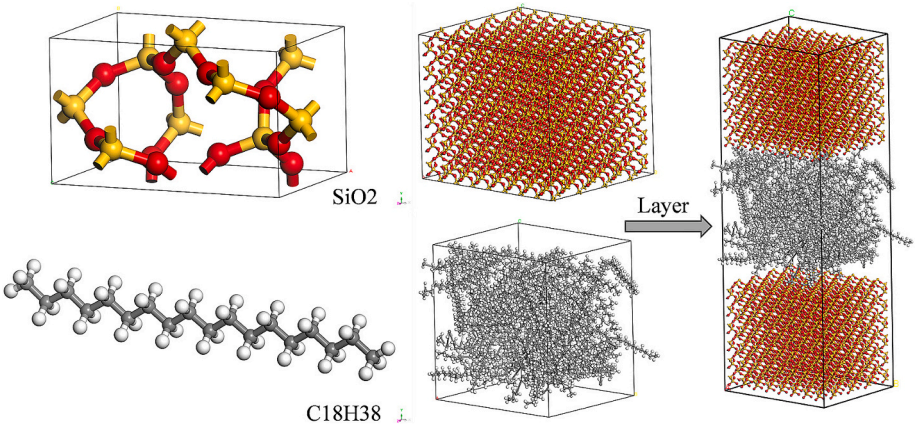


Fig. 1. N-octadecane/silica phase change microcapsule modeling flow chart.

improve the heat transfer efficiency of phase change materials [9], and facilitate the use, storage and transportation of phase change materials [10]. The size of these solid particles is generally between 1 and 1000 μm [11]. At present, there are more than 200 kinds of preparation methods for phase change microcapsules [12], and they are widely used in many fields such as food and medicine production [13–16], solar energy storage [17,18], textiles [19], building energy conservation [20], and micro-channel heat transfer [21–24].

Yan et al. [25] prepared n-eicosane core/polyurea shell phase change microcapsules (MPCM) using n-eicosane water-in-oil emulsion emulsified with lignin sulfonate, showing good dispersion, uniform particle size, high phase change enthalpy (209.8 J /g), high encapsulation rate (82.9 %), excellent thermal cycle stability and extremely low leakage rate. Demonstrated excellent heat storage and thermal management capabilities. Rao et al. [26] studied the thermal energy storage mechanism of nanoencapsulated and nanoparticle reinforced phase change materials (PCM) through molecular dynamics simulation. The results show that shell thickness and nanoparticle diameter significantly affect the self-diffusion coefficient, radial distribution function, end-to-end distance and bond torsion Angle distribution of PCM. Optimizing the shell thickness to an appropriate level can significantly enhance mass diffusion efficiency and heat energy storage capacity. However, shells that are either excessively thick or overly thin may impede the effective encapsulation of PCM. Wen et al. [27] prepared mononuclear phase change microcapsules (S − PCM) and multi-nuclear phase change microcapsules (M − PCM) with paraffin wax (melting point 37 °C) as the core and crosslinked chitosan as the wall. S − PCM latent heat is 61.4 mJ/mg, M − PCM latent heat is 50.1 mJ/mg. Both showed good stability without paraffin leakage. Qiu et al. [28] constructed two hybrid structures of carbon nanotubes and polyurethane, and found through non-equilibrium molecular dynamics (NEMD) simulation that the interface thermal conductance significantly increased with the increase of the length of single-wall carbon nanotubes and the number of polyurethane molecular chains. Deng et al. [29] prepared paraffin (Pa) @silica (SiO2) microcapsules with different core/shell ratios by interphase condensation polymerization method, and embedded them in addition solidified liquid silicone rubber (ALSR) to prepare the ALSR/ Pa@SiO2 phase change composite. The thermal storage performance of the composite can reach 90.7 J /g, and the enthalpy of phase transition increases with the increase of core/shell ratio. Due to the interaction of the chemical interface, the mechanical strength of the composite material is improved, and it can be used as an electronic packaging material for thermal management and temperature regulation.

The research on phase change microcapsules, on the one hand, focuses on their preparation and application, using experimental methods to prepare phase change microcapsules and using measuring instruments to characterize their physical characteristics and thermal properties [30]. On the other hand, some modified materials or structures are simulated in terms of thermal properties through molecular dynamics, such as thermal conductivity, enthalpy value, and some microscopic indicators, such as phonon transport, radial distribution function, etc. However, there are relatively few studies on the heat transfer process of individual microcapsules. This paper used SiO2 as the shell material and n-octadecane as the core material. LAMMPS software was used to simulate its molecular dynamics. In this study, the temperature distribution characteristics of core-shell structures under different temperature differences and core/shell ratios during heat transfer were systematically discussed.

2. Model and methodology

2.1. Physical model

In this study, the modeling was conducted utilizing the Materials Studio software. The lattice constants of the SiO2 crystal were precisely determined as a = 8.509 Å, b = 4.913 Å, and c = 5.405 Å. To construct larger supercells, a series of cell expansion operations were performed, resulting in modular supercells with dimensions of a = 34.038 Å, b = 39.304 Å, and c = 32.431 Å, corresponding to a 4 × 8 × 6 expansion. The dimensions of the fixed cells in the x and y directions were maintained at 34.038 Å × 39.304 Å, while the dimension in the z direction was set to 40.6 Å. The spherical structure of the phase change microcapsule is simplified to a three-layer cuboid, which facilitates the application of periodic boundary conditions and greatly simplifies the calculation process. Although the surface area to volume ratio of a cuboid model is usually higher than that of a spherical model. This may lead to higher heat loss and faster temperature changes, especially during the heat transfer process. This difference may affect the accuracy of the simulation results. However, this paper mainly studies the heat transfer process in the direction perpendicular to the core-shell and the variation law of the core-shell ratio. The above-mentioned differences are acceptable to a certain extent. Specifically, the middle layer represents the core material undergoing phase change, whereas the upper and lower layers correspond to the shell material. Layered modeling was applied to both SiO2 and paraffin. The detailed modeling procedure is illustrated in Fig. 1.

2.2. Simulation method

In molecular dynamics simulations, potential functions describe the forces between atoms, and different types of substances require different types of potential functions. For the composite material system, the composite potential function is set to calculate, cvff potential function is selected for n-octadecane, tersoff potential function for SiO2 [31], and Lennard-Jones (L-J) potential function is selected for the interaction

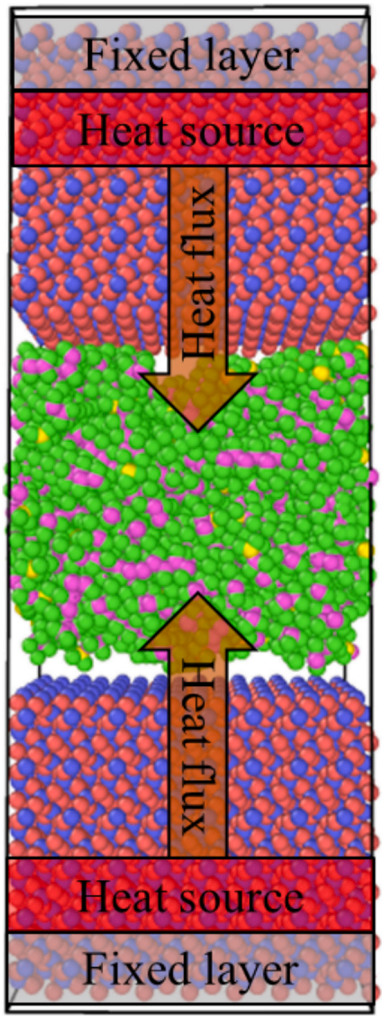


Fig. 2. Heating diagram of system under Langevin temperature control method.

between the two:

$$E(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \tag{1}$$

Where E is the potential energy, ε is the depth of the potential well, σ is the equilibrium distance when the potential energy of the interaction is zero, and r is the interatomic distance, which is less than the truncation radius. The mixing rule can be used to predict the potential parameters between two distinct atoms if the L-J potential parameters between atoms of the same species are known:

$$\sigma_{ij} = \frac{1}{2}\left(\sigma_{ii} + \sigma_{jj}\right) \tag{2}$$

$$\varepsilon_{ij} = \sqrt{\varepsilon_{ii} + \varepsilon_{jj}} \tag{3}$$

In this study, the Langevin thermostat [32] was employed in the LAMMPS simulations to control the temperature of the system. The Langevin thermostat is a widely used method in molecular dynamics simulations, which introduces a stochastic force and a damping term to mimic the interaction between the system and a heat bath. The Langevin equation can be expressed as:

$$m\frac{dv}{dt} = \mathbf{F}_c + \mathbf{F}_f + \mathbf{F}_r \tag{4}$$

where m is the mass of the particle, $\mathbf{v}$ is the velocity, $\mathbf{F}_c$ is the conservative force derived from inter-particle interactions, $\mathbf{F}_f$ is the frictional force, and $\mathbf{F}_r$ is the random force. The frictional force is given by:

$$\mathbf{F}_f = -\frac{m}{\tau}\mathbf{v} \tag{5}$$

where τ is the damping parameter, which controls the strength of the frictional force.

$$\mathbf{F}_r \propto \sqrt{\frac{k_B T m}{\Delta t \tau}} \tag{6}$$

The random force $\mathbf{F}_r$ is proportional to the above equation, where k_B is the Boltzmann constant, T is the target temperature, and Δt is the time step of the simulation. This combination of forces ensures that the system reaches and maintains the desired temperature while also introducing thermal fluctuations consistent with the canonical ensemble.

Table 1
Simulation of different temperature difference and core/shell ratio.

Simulated conditions	Relaxation temperature	Heat source temperature (K)	ΔT (K)	Core/ shell ratio
1	273	323	50	0.68
2	273	373	100	0.68
3	273	423	150	0.68
4	273	423	150	0.86
5	273	423	150	1.18

2.3. Simulation process

The open-source program LAMMPS was utilized to perform all of the computations in this work. Along the Z axis, the system was separated into 20 layers, with the two atomic layers at either end being fixed. The heat source was simultaneously set to the atom layer at both ends, with the exception of the fixed layer. The system's heating diagram under the Langevin temperature control approach is displayed in Fig. 2. Firstly, energy minimization and temperature initialization of the whole simulation system are carried out to obtain a stable structure. Then, in order to keep the system at a stable initial temperature, the Nosé-Hoover thermostat is used for relaxation. Relaxation simulations at 80 ps were performed under both the isothermal-isobaric ensemble (NPT) and the canonical ensemble (NVT). After the stable state is reached, the Langevin thermostat is performed under the microcanonical ensemble (NVE) to calculate the temperature and energy of the system. Due to the presence of a temperature gradient within the system during the heat transfer process, the Langevin thermostat exhibits greater flexibility in handling non-equilibrium state simulations and is capable of more accurately simulating the interaction between the system and its surroundings.

The temperature of the heat source was changed to provide a range of temperature disparities in order to model the temperature distribution of a system under varied thermal conditions. Similarly, the thickness of the fixed layer was used to control the thickness of the shell; a thicker fixed layer is associated with a thinner SiO_2 layer and a higher core/shell ratio. In this study, three temperature differences were set: 50 K, 100 K, and 150 K. Additionally, three core/shell ratios were established: 0.68, 0.86, and 1.18. Comparative analyses were conducted among these conditions. All simulated conditions are summarized in Table 1. The simulations were run for 300 ps, with results output every 0.5 ps to facilitate the selection of data for post-processing analysis.

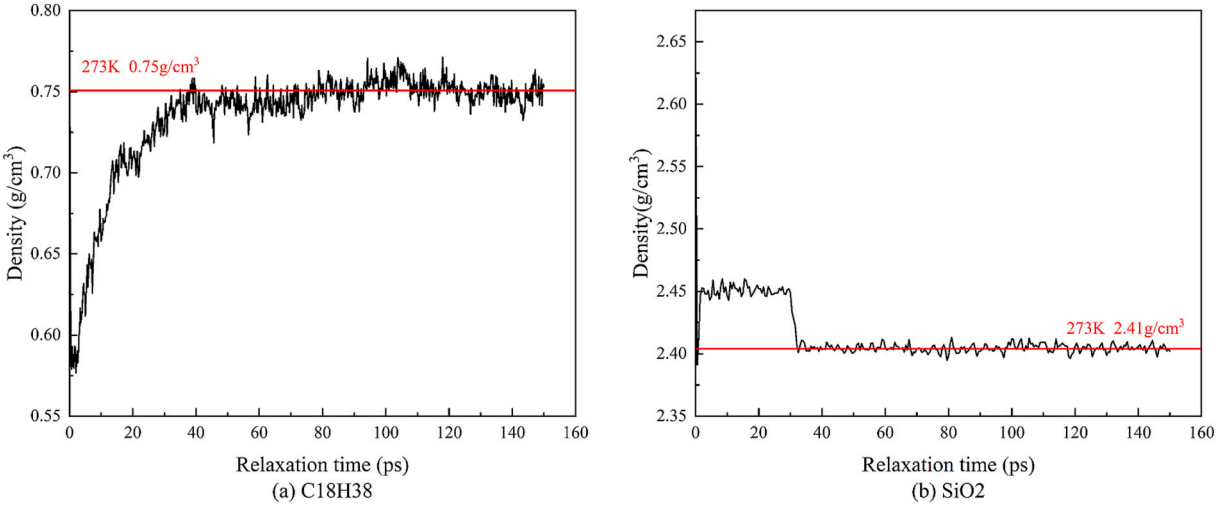


Fig. 3. Density change with relaxation time at 273 K temperature.

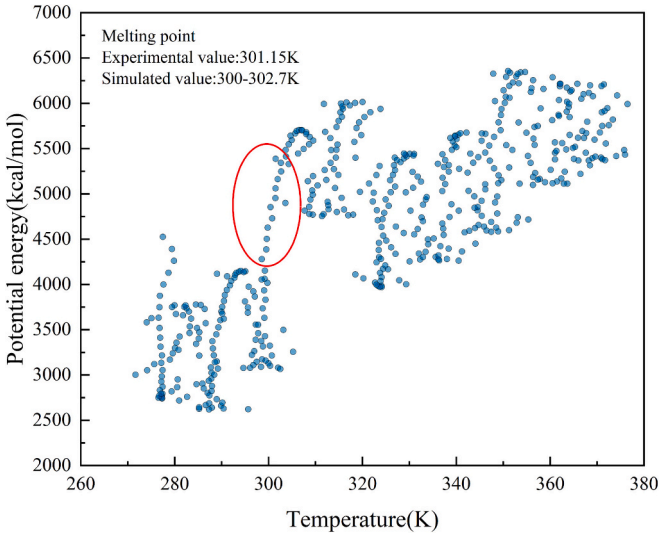


Fig. 4. The change of potential energy with the heating process.

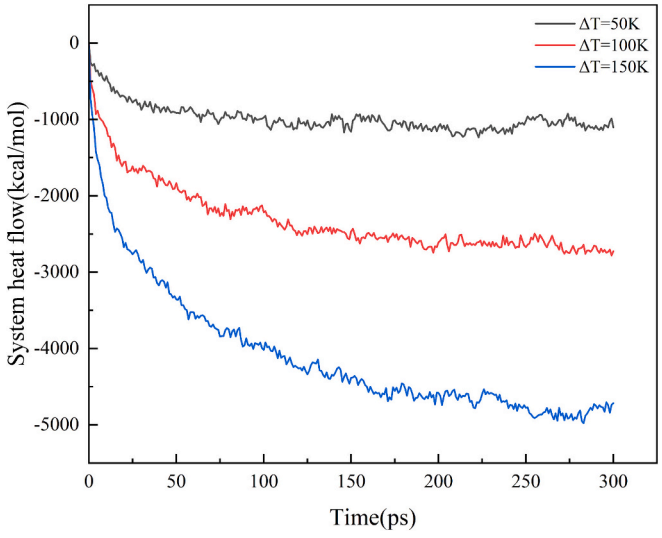


Fig. 5. System heat flow diagram under different temperature difference conditions.

3. Results and discussions

3.1. Model verification

3.1.1. Density

To validate the rationality of the model and the appropriateness of the selected potential function, NPT relaxation simulations were conducted on C18H38 and SiO2 at a temperature of 273 K. The density variations of these materials over time were monitored throughout the simulation process. As seen in Fig. 3, the density of C18H38 steadily stabilized throughout the course of the 160 ps simulation period, eventually arriving at 0.75 g/cm^3. This value exhibits a relative error of 3.47 % when compared to the experimental density of 0.777 g/cm^3. Similarly, the density of SiO2 stabilized at 2.41 g/cm^3, which corresponds to a relative error of 3.89 % relative to its experimental density of 2.31 g/cm^3. Given that these relative errors fall within the acceptable range, it can be concluded that the model and the potential function employed in this study are both rational and applicable. Additionally, the density tends to stabilize around 40 ps, therefore the simulation calculation's relaxation time of 80 ps is sufficient to bring the system to stability.

The reasons for the existence of errors come from multiple aspects. For the selection of the potential function, the van der Waals force parameter in the CVFF potential function directly affects the intermolecular distance in the system. The key length and key Angle parameters in the Tersoff potential function also have a direct impact on the density calculation. These parameters have not been verified through sufficient experiments or high-precision calculations, which may lead to deviations between the simulation results and the experimental values. Furthermore, a finite-sized simulation system may introduce boundary effects, leading to deviations in intermolecular interactions and thereby affecting density calculations.

3.1.2. Energy

The system is heated up from 273 K to 373 K, and the scatter diagram of potential energy changing with temperature is shown in Fig. 4. It can be observed that there is a significant surge of potential energy in the 300-303 K temperature range. During the melting process, the paraffin molecules gradually lose their fixed location in the solid state. Once they enter the liquid state, the intermolecular distance grows and the intermolecular force decreases, which raises the potential energy. The magnitude of the intermolecular contact force that the paraffin molecules must overcome during the phase transition process is reflected in

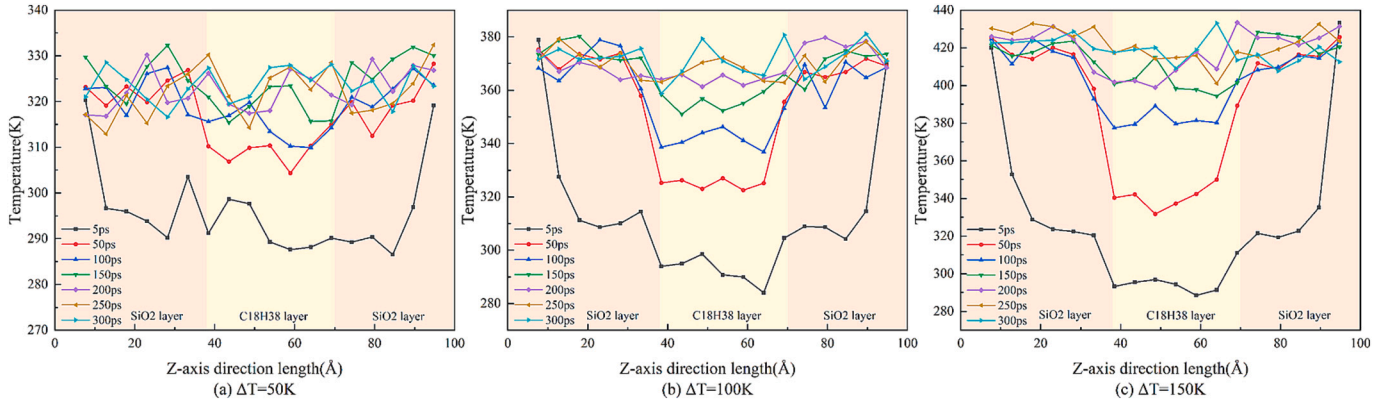


Fig. 6. The temperature gradient changes with different temperature differences.

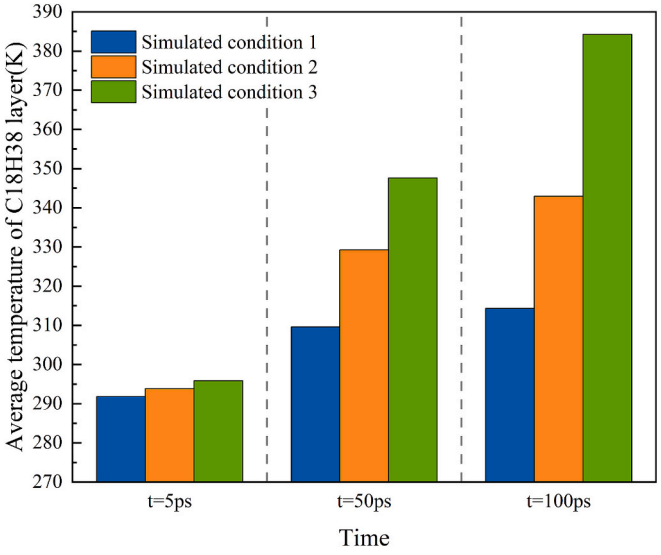


Fig. 7. Average temperature of layer C18H38 at different times.

this potential energy surge, which also indicates the presence of latent heat of the phase transition. Therefore, 300 K–303 K can be considered as the temperature range in which the phase transition occurs in the system. This is consistent with the experimental melting point of n-octadecane [33]. During the simulation of different working conditions, the heat flow values in the system under three temperature differences are output respectively, as shown in Fig. 5. The greater the temperature difference, the greater the heat flow value, and the greater the temperature difference between the external temperature and the microcapsule, the greater the heat flow generated in each microcapsule system. At the same time, it is proved that the simulation time of 300 ps is enough for the three temperature difference conditions to reach stability, that is, at 300 ps, the temperature of the two substances in the system has reached the heat source temperature.

3.2. *Influence of the temperature difference*

In this study, the system is divided into 20 equal layers along the *Z*-axis, and the temperature of each layer is monitored. After excluding the points with special temperature values in the fixed layer and the heat source layer, the temperature data of seven specific time points, namely 5 ps, 50 ps, 100 ps, 150 ps, 200 ps, 250 ps and 300 ps, were extracted respectively, and then the corresponding temperature gradient diagram was drawn, as shown in Fig. 6. Where (a), (b) and (c) respectively correspond to simulated conditions 1, 2, and 3 in Table 1. The heat of the heat source is quickly transported to the C18H38 layer at a temperature differential of 50 K, as shown in Fig. 6(a). The phase transition temperature is attained at 50 ps, and after 150 ps, the system's temperature is essentially stable at 323 K.

By comparing Fig. 6(b) and (c), it can be seen that the SiO2 layer has a faster heat transfer rate with its good thermal conductivity, and basically reaches the heat source temperature at 50 ps. In the middle of the C18H38 layer, with the gradual increase of temperature, the heating rate gradually slowed down. This suggests that the phase change microcapsule with shell and core structure may rapidly transmit external heat to the core thanks to the shell's superior thermal conductivity. The heating rate of the PCM at the core is positively correlated with the temperature difference between the PCM and the ambient temperature. Specifically, as the temperature difference increases, the heating rate also rises accordingly.

The average temperature of the C18H38 layer in Fig. 6 was calculated, and the values of 5 ps, 50 ps and 100 ps were selected to draw the histogram as shown in Fig. 7. The influence of temperature difference on the temperature rise of C18H38 layer is not particularly significant at the moment of 5 ps. However, with the deepening of heat transfer, the temperature gap between the three working conditions at 50 ps and 100 ps becomes larger and larger. This indicates that when the temperature rises to close to the heat source temperature, the temperature rise rate also slows down.

In order to more intuitively reflect the temperature distribution in the system, the YOZ plane is divided into 5 × 20 small blocks, the average temperature of atoms contained in each block can be calculated to draw the temperature contour plot. When the temperature difference between the heat source and the system temperature is 150 K, the temperature cloud image at 5 ps, 50 ps, 100 ps and 150 ps is shown in Fig. 8. Since the atomic momentum of the fixed layer is set to zero in the simulation, the temperature obtained from the momentum calculation is also zero. The outer block adjacent to the heat source layer contains a portion of atoms from the fixed layer. This leads to a situation in which the average temperature of the block is lower than 273 K in the calculation process. The temperature contour plot visually presents the temperature distribution of the entire system during the heat transfer process, but it is difficult to depict parameters such as latent heat release during the phase change process.

It can be more vividly observed from the temperature contour plot that at 5 ps, the temperature of the entire system is relatively low. The heat source has only a minimal effect on the SiO2 layer, while the central C18H38 layer reaches 273 K after relaxation. At 50 ps, the SiO2 layer has all risen to the heat source temperature, and the intermediate layer is all higher than the melting point temperature of C18H38. At 100 ps, the temperature rises further, and at 150 ps, the system has basically reached the heat source temperature of 423 K.

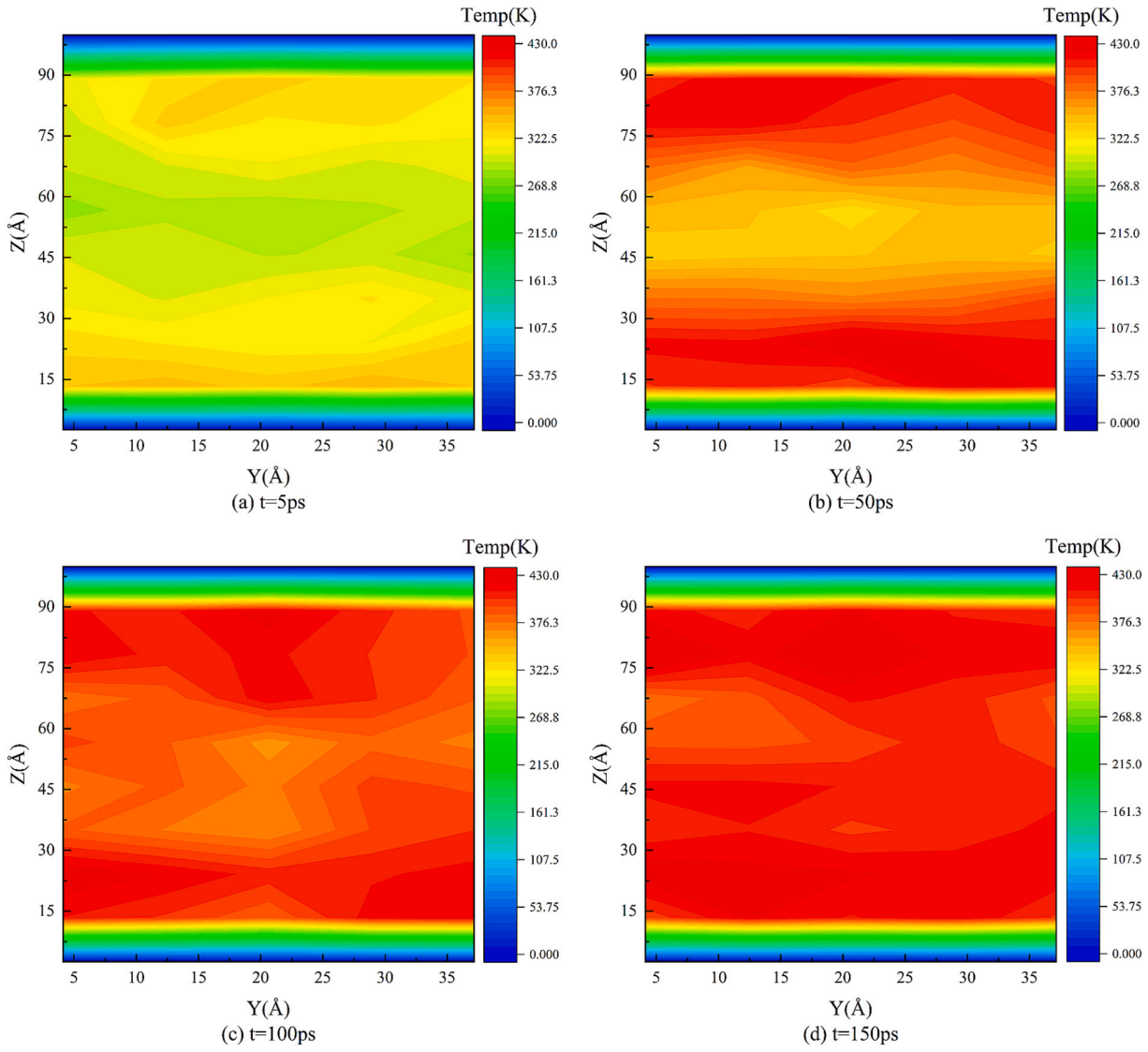


Fig. 8. Temperature contour plot at different times, ΔT = 150 K.

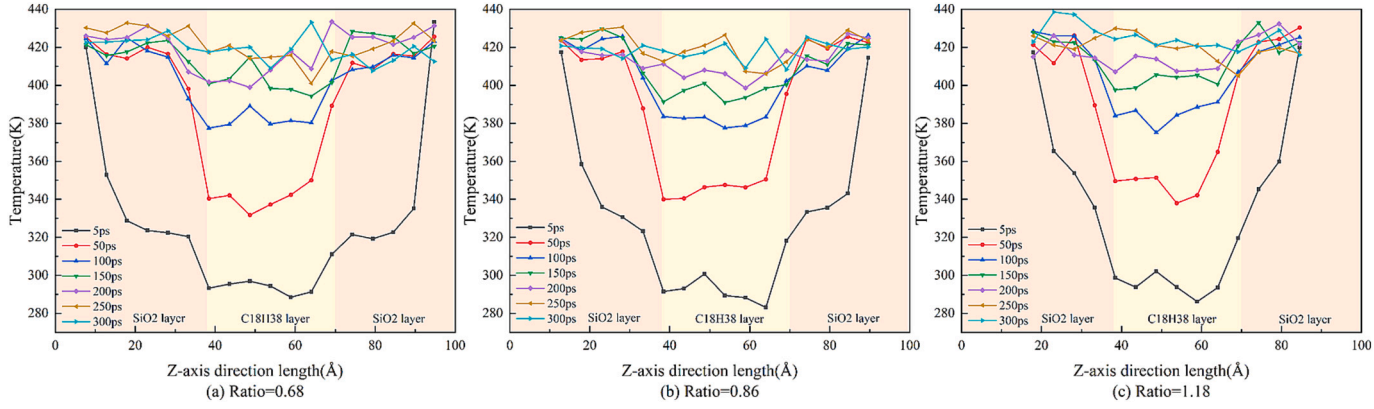


Fig. 9. Temperature gradients of different core/shell ratios.

3.3. *Influence of core/shell ratio*

After discussing the influence of temperature difference on heat transfer, this section focuses on the influence of different core/shell ratio on heat transfer process of phase change microcapsules. Similar to the above, Fig. 9 shows the temperature gradients of the system with core/shell ratios of 0.68, 0.86 and 1.18. Where Fig. 9 (a), (b) and (c) respectively correspond to simulated conditions 3, 4, and 5 in Table 1.

In this study, altering the fixed layer's thickness modifies the core to shell ratio. The matching SiO2 layer is thinner and the core to shell ratio

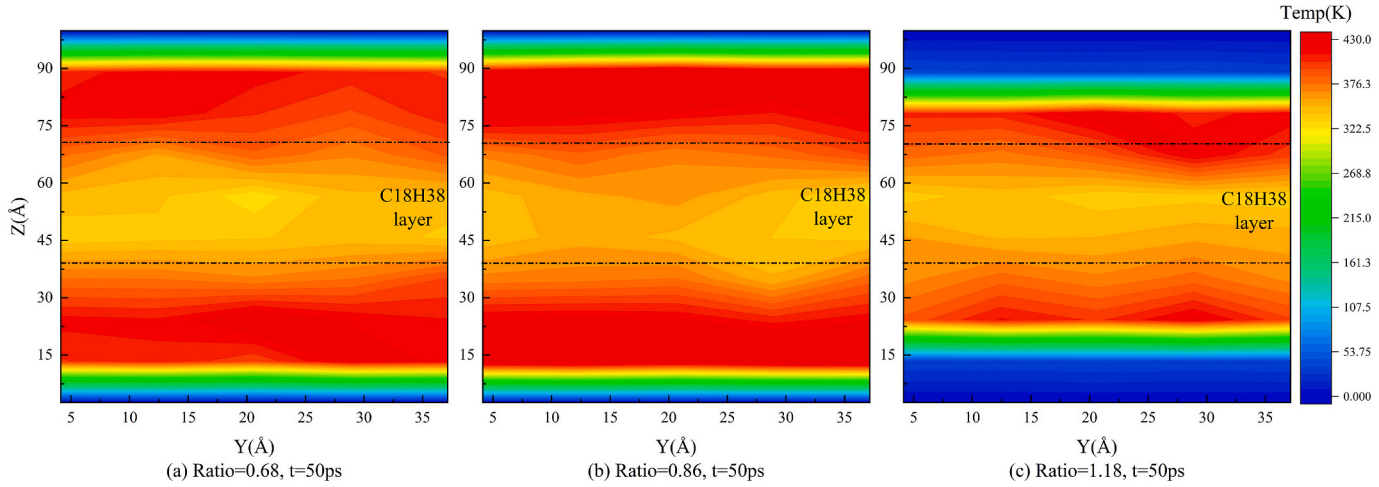


Fig. 10. Temperature contour plot of different core/shell ratios at 50 ps.

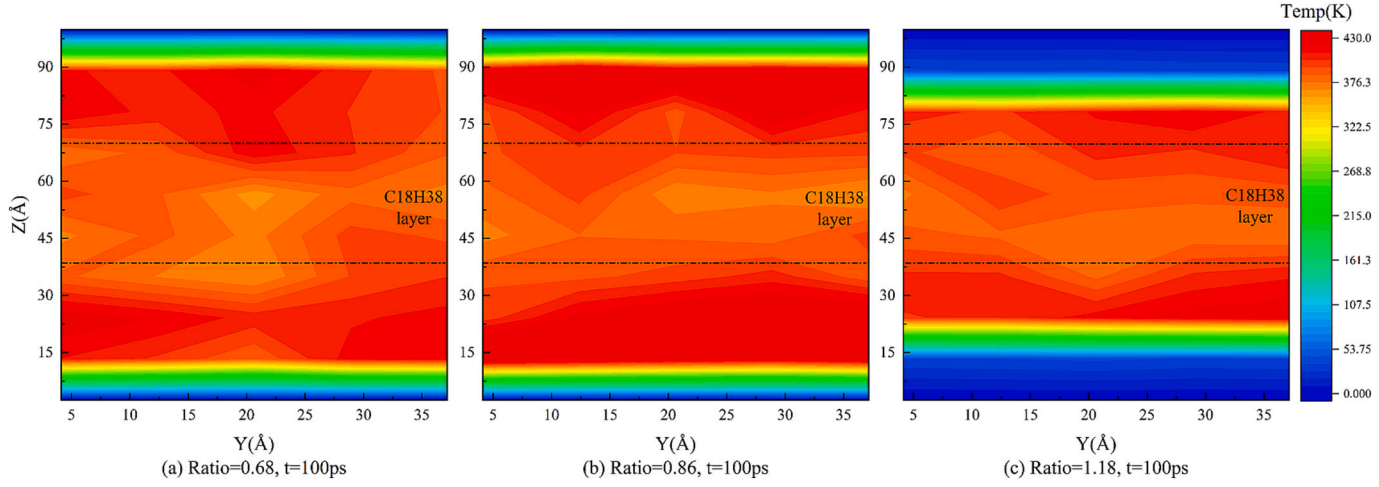


Fig. 11. Temperature contour plot of different core/shell ratios at 100 ps.

is bigger when the fixed layer area is set to a thicker value. Comparing the temperature points at the same time in Fig. 9 (a), (b) and (c), it can be observed that the temperature point distribution of C18H38 layer will slightly increase with the increase of core/shell ratio. It is shown that reducing the thickness of the shell can indeed increase the heating rate of the PCM. However, because the thermal conductivity of the shell material is obviously higher than that of the phase change material, the shell thickness is not the main factor affecting the thermal resistance of the entire phase change microcapsule. Reducing shell thickness to increase heat transfer rate has minimal impact.

The temperature contour plot of different core/shell ratios at 50 ps is shown in Fig. 10. Similarly, the temperature contour plot at 100 ps is shown in Fig. 11. There is essentially no temperature differential in the Y-axis direction, and the heat transmission effect is good as nearly all of the SiO2 layer reaches the heat source temperature. The consistent and mild temperature gradient layer is visible in the *Z*-axis direction. For the C18H38 layer, the thermal phase transition increases the disorder of atomic motion and the frequency of collision. The temperature distribution along the Y-axis is inconsistent, and the temperature stratification is more gentle and orderly when the core/shell ratio is larger. Especially in Fig. 11, it can be seen more obviously that the larger the core/shell ratio, the higher the overall temperature of the C18H38 layer.

4. Conclusion

In this paper, a three-layer model of SiO2 and C18H38 is constructed. Langevin temperature control method was used to simulate the heat transfer effect of each layer of phase change microcapsule when heated by external temperature. The qtemperature distribution in the system at different times is presented mainly in the form of temperature gradient and temperature contour plot, and analyzed from two perspectives of temperature differences and different core/shell ratios. The main conclusions are summarized as follows:

(1) During the heat transfer process of phase change microcapsules, the influence of temperature difference on the temperature rise of the C18H38 layer gradually emerges over time. The reduction in temperature difference results in a diminished heat conduction rate, which aligns with Fourier's law. If the heat storage characteristics of phase change materials are to be utilized, it is possible to consider increasing the temperature difference between the phase change microcapsules and the heat exchange medium in the application.

(2) For the phase change microcapsules with shell-core structure, the superior thermal conductivity of the silica shell layer enables it to quickly reach the heat source temperature, thereby promoting the rapid transfer of heat to the interior. Therefore, optimizing

the thermal conductivity of the shell material is one of the key factors for improving the heat transfer efficiency of phase change microcapsules.

(3) The increase of core/shell ratio has a certain influence on the heat transfer rate of the entire system. The larger the core/shell ratio is, the smaller the proportion of the shell layer decreases, which is more conducive to the rapid transfer of heat to the core-layer phase change material. However, the thermal conductivity of the shell material is much greater than that of the phase change material, and it is not the main factor for the thermal resistance of the entire microcapsule system. Therefore, emphasis should be placed on the selection and optimization of shell materials rather than simply relying on the core-shell ratio to change the heat transfer rate of microcapsules.

CRediT authorship contribution statement

Juhui Chen: Methodology, Conceptualization. **Ran An:** Writing – original draft, Software. **Dan Li:** Formal analysis. **Michael Zhuravkov:** Writing – review & editing. **Siarhei Lapatsin:** Writing – review & editing. **Wenrui Jiang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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